

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Which of the following is an adaptation that enhances the uptake of water and minerals by roots?
 - mycorrhizae
 - pumping through plasmodesmata
 - active uptake by vessel elements
 - rhythmic contractions by cells in the root cortex
- Which structure or compartment is part of the symplast?
 - the interior of a vessel element
 - the interior of a sieve tube
 - the cell wall of a mesophyll cell
 - an extracellular air space
- Movement of phloem sap from a source to a sink
 - occurs through the apoplast of sieve-tube elements.
 - depends ultimately on the activity of proton pumps.
 - depends on tension, or negative pressure potential.
 - results mainly from diffusion.

Levels 3-4: Applying/Analyzing

- Photosynthesis ceases when leaves wilt, mainly because
 - the chlorophyll in wilting leaves is degraded.
 - accumulation of CO_2 in the leaf inhibits enzymes.
 - stomata close, preventing CO_2 from entering the leaf.
 - photolysis, the water-splitting step of photosynthesis, cannot occur when there is a water deficiency.
- What would enhance water uptake by a plant cell?
 - decreasing the Ψ of the surrounding solution
 - positive pressure on the surrounding solution
 - the loss of solutes from the cell
 - increasing the Ψ of the cytoplasm
- A plant cell with a Ψ_s of -0.65 MPa maintains a constant volume when bathed in a solution that has a Ψ_s of -0.30 MPa and is in an open container. The cell has a
 - Ψ_p of $+0.65$ MPa.
 - Ψ of -0.65 MPa.
 - Ψ_p of $+0.35$ MPa.
 - Ψ_p of 0 MPa.
- Compared with a cell with few aquaporin proteins in its membrane, a cell containing many aquaporin proteins will
 - have a faster rate of osmosis.
 - have a lower water potential.
 - have a higher water potential.
 - accumulate water by active transport.
- Which of the following would tend to increase transpiration?
 - spiny leaves
 - sunken stomata
 - a thicker cuticle
 - higher stomatal density

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Large brown algae called kelps can grow as tall as 25 m. Kelps consist of a holdfast anchored

to the ocean floor, blades that float at the surface and collect light, and a long stalk connecting the blades to the holdfast (see Figure 28.13). Specialized cells in the stalk, although nonvascular, can transport sugar. Suggest a reason why these structures analogous to sieve-tube elements might have evolved in kelps.

10. SCIENTIFIC INQUIRY • INTERPRET THE DATA

A Minnesota gardener notes that the plants immediately bordering a walkway are stunted compared with those farther away. Suspecting that the soil near the walkway may be contaminated from salt added to the walkway in winter, the gardener tests the soil. The composition of the soil near the walkway is identical to that farther away except that it contains an additional 50 mM NaCl. Assuming that the NaCl is completely ionized, calculate how much it will lower the solute potential of the soil at 20°C using the *solute potential equation*:

$$\Psi_s = -iCRT$$

where i is the ionization constant (2 for NaCl), C is the molar concentration (in mol/L), R is the pressure constant [$R = 0.00831 \text{ (L} \cdot \text{MPa)} / (\text{mol} \cdot \text{K})$], and T is the temperature in Kelvin ($273 + ^\circ\text{C}$).

How would this change in the solute potential of the soil affect the water potential of the soil? In what way would the change in the water potential of the soil affect the movement of water in or out of the roots?

- SCIENTIFIC INQUIRY** Cotton plants wilt within a few hours of flooding of their roots. The flooding leads to low-oxygen conditions, increases in cytosolic Ca^{2+} concentration, and decreases in cytosolic pH. Suggest a hypothesis to explain how flooding leads to wilting.

- WRITE ABOUT A THEME: ORGANIZATION** Natural selection has led to changes in the architecture of plants that enable them to photosynthesize more efficiently in the ecological niches they occupy. In a short essay (100–150 words), explain how shoot architecture enhances photosynthesis.

13. SYNTHESIZE YOUR KNOWLEDGE



Imagine yourself as a water molecule in the soil solution of a forest. In a short essay (100–150 words), explain what pathways and what forces would be necessary to carry you to the leaves of these trees.

For selected answers, see Appendix A.